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Part 1: Connection Time Experiment

Paste your screenshot of the connection benchmark output below:

```
Testing Mutual TLS (port 8884)...
C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\mtls_benchmark.py:112: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
n
client = mqtt.Client(
Trial 1: 28.3 ms
Trial 2: 24.0 ms
Trial 3: 19.7 ms
Trial 4: 31.2 ms
Trial 5: 23.4 ms
Trial 6: 19.5 ms
Trial 7: 14.3 ms
Trial 8: 19.2 ms
Trial 9: 49.3 ms
Trial 10: 14.0 ms
Trial 11: 27.0 ms
Trial 12: 36.5 ms
Trial 13: 27.5 ms
Trial 14: 17.0 ms
Trial 15: 12.1 ms
Trial 16: 16.5 ms
Trial 17: 19.9 ms
Trial 18: 35.6 ms
Trial 19: 20.1 ms
Trial 20: 31.5 ms

=====
Connection Time Comparison
=====
Trials: 20

One-Way TLS:
Average: 21.8 ms
Min: 7.6 ms | Max: 29.0 ms

Mutual TLS:
Average: 24.3 ms
Min: 12.1 ms | Max: 49.3 ms

Overhead: +2.5 ms (+11.5%)
=====
```

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Connection Time Comparison
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Average: 24.3 ms
Min: 12.1 ms | Max: 49.3 ms

Overhead: +2.5 ms (+11.5%)
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Connection Time Results

Metric	One-Way TLS	Mutual TLS
Average connection time	21.8 ms	24.3 ms
Min connection time	7.6 ms	12.1 ms
Max connection time	29.0 ms	49.3 ms
Overhead	--	+ 2.5 ms (+ 11.5 %)

Part 2: Message Latency Experiment

Paste your screenshot of the latency benchmark output below:

```
=====
mTLS Benchmark: Message Latency
=====
Sending 50 messages for each mode...

Testing One-Way TLS (port 8883)...
C:\Users\gbem1\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\mtls_benchmark.py:238: DeprecationWarning: Callback API version 1 is deprecated, update to latest versio
n
  sub_client = mqtt.Client(client_id=f"bench-sub-{port}", **MQTT_CLIENT_ARGS)
C:\Users\gbem1\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\mtls_benchmark.py:261: DeprecationWarning: Callback API version 1 is deprecated, update to latest versio
n
  pub_client = mqtt.Client(client_id=f"bench-pub-{port}", **MQTT_CLIENT_ARGS)
Connected. Publishing 50 messages...
Average latency: 1.7 ms

Testing Mutual TLS (port 8884)...
Connected. Publishing 50 messages...
Average latency: 1.6 ms

=====
Message Latency Comparison
=====
Messages: 50

One-Way TLS:
Average: 1.7 ms
Min: 1.0 ms | Max: 4.3 ms

Mutual TLS:
Average: 1.6 ms
Min: 1.1 ms | Max: 2.3 ms

Overhead: -0.1 ms (-5.5%)
=====
```

Message Latency Results

Metric	One-Way TLS	Mutual TLS
Average latency	1.7 ms	1.6 ms
Min latency	1.0 ms	1.1 ms
Max latency	4.3 ms	2.3 ms
Overhead	--	- 0.1 ms (-5.5 %)

Part 3: Analysis

1. Connection Overhead

How many extra milliseconds does mTLS add to connection setup?

Your answer: 2.5ms

The Grand Marina's devices connect once and stay connected for hours. Does this one-time cost matter?

Your answer: No, it's like 25 seconds once every day!

2. Message Latency

How many extra milliseconds does mTLS add per message?

Your answer: it is actually lower or basically the same, (-0.1 ms)

Sensor readings are sent every 5 seconds (5,000 ms). What percentage of that interval is the mTLS overhead?

Your answer: -5.5%